

Business Entity Resolution

Amazon ML Challenge 2026 — what the problem is, what we are given, what we must deliver, the rules we must not break, and how we plan to win. All data figures were measured on the provided files.

1.73M test Source-1 entities to resolve	≈10M test Source-2 + Source-3 records to search	F0.5 macro, per entity (precision ×2)	France 15% of test, absent from training
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1. The problem in plain words

Three independent systems each hold records about businesses. They share **no common ID**, and the same business is written differently in each one (typos, abbreviations, reordered addresses, names in Indian scripts, missing fields). **Source 1 is the clean, deduplicated master list.** For every Source 1 business in the test set, we must list every Source 2 and Source 3 record that is the same real-world business. A business can have zero, one, or many such records.

```
S1-496331216  Shree Best Business Private Limited
                406, Manas Nagar Colony, Ankit Apartment, Lucknow, Uttar Pradesh
├─ S2-198865808  <same name in Devanagari script> | 406, MANAS NAGAR COLONY, ..., LUCKNOW
├─ S3-43022101   Shree Best Business Limited Service | #851 406, UP, Lucknow
└─ S3-247018126  Shree Best Business Private-Ltd    | Door No 851 406, Lucknow, UP

S1-299182413  East Advanced Cohen LLC | Pulaski, TN, 1189 Sumac Road
├─ S2-957790650  East Advanced Cohen Llc | 1189 SUMAC RD, PLUASKI, TN
└─ S2-259918198  TAVOFLUX                | 1189 SUMAC ROAD, TN, PLUASKI  <- other trade name!
```

Real training examples: one S1 entity and its true S2/S3 matches.

2. What we are given (inputs)

All files are **UTF-8, tab-separated (.tsv)**. Each source file has four columns: `entity_id` (prefix S1-/S2-/S3- plus a random number), `business_name`, `business_address`, and `country`. The ground-truth file has `source1_entity_id` and `matched_entity_ids` (a comma-separated list, empty when there is no match).

File	Rows	Country mix	Role
train_source1.tsv	2,206,821	US 60% · India 40%	Reference entities (labelled)
train_source2.tsv	5,034,616	US 3.02M · India 2.02M	Records to match
train_source3.tsv	5,285,603	US 3.17M · India 2.12M	Records to match
train_ground_truth.tsv	2,206,821	—	The answers for training
test_source1.tsv	1,732,544	US 38% · India 47% · France 15%	Entities we must answer for
test_source2.tsv	4,887,273	US 1.87M · India 2.31M · France 0.70M	Search pool
test_source3.tsv	5,082,316	US 1.95M · India 2.41M · France 0.73M	Search pool

Also provided: `utils/validate_submission.py` (format checker), `Documentation_template.md` (methodology write-up to fill in), and a README identical to the problem statement. Total size is about 2.4 GB.

3. What the data tells us (EDA)

Finding	Evidence	What it means for us
Every S2/S3 record belongs to at most one S1	7,638,365 links; every linked ID is used exactly once	We can enforce a one-owner assignment, which is a strong precision tool.

Finding	Evidence	What it means for us
It is one-to-many	Mean 3.46 matches per S1; most have 2–6 (max 11)	Returning only the single best match leaves recall on the table.
Singletons are rare	5.6% of S1 have no match	A wrong match on a singleton turns 1.0 into 0.0.
About a quarter of S2/S3 match nothing	Unmatched: S2 26.6%, S3 25.4%	Distractors are everywhere, so a similar name alone is not proof.
No cross-country matches	0 of 7.6M links	Block within each country label (lossless).
Test pool is relatively larger	S2+S3 per S1: train 4.68, test 5.75	Expect a harder precision problem on test than on validation.
S1 is clean; S2/S3 are noisy	S2/S3: ≈3% empty addresses; ≈40% of Indian records in Indic scripts (9 scripts)	We need normalisation and transliteration.
Postal codes are basically absent	India PIN: 0; US ZIP: under 0.01%	We can't block on ZIP/PIN; use city, street and house number.
IDs carry no information	No train/test ID overlap; random numbers	Never use ID values as features.
Parsing traps	≈800 test rows with quoted fields like ""ehpad Club SAS"; ≈100 NA/null values	Load with quoting off and NA-conversion off, then assert row counts.

Noise patterns seen in true matches

- **Names:** typos, leetspeak (*Sh0shana*, *5hoshana*, *Carg0*), random accents (*Émpire*), legal suffix added, dropped or reworded (*Pvt Ltd / Private Limited*), filler words (*Center*, *Services*, *Group*), word reordering, junk prefixes (*--*, *<<*, *#*), domains and handles (*pediatricmedicine.com*), aliases (*X dba Y*, *X fka Y*), a completely different trade name at the same address, and whole names transliterated into Indic scripts.
- **Addresses:** components reordered, *Rd/Road*, *TN/Tennessee*, *UP/Uttar Pradesh*, house number off by a little (*2046/2048*) or given as a range (*120-122*), injected *#*, *Door No*, *HN*, *PO Box*, neighbouring or renamed cities (*Gurgaon/Gurugram*), city typos, missing parts, or no address at all.
- **France (test only):** SARL/SAS/SASU/EURL/SCI legal forms, *R./AV./BD.* street abbreviations, region vs département (*Hauts-de-France / Nord*), accents on and off, few cities, and very generic names (*Association des Coeur*, *Clinique Jeanne*).

4. What we must produce (outputs)

(a) **output/matching_results.tsv** — the only scored file, uploaded to the leaderboard portal.

```
source1_entity_id matched_entity_ids
S1-00001 S2-00047,S2-00193,S3-00812
S1-00002 S3-00004
S1-00003
^ empty = we believe this entity has no match
```

(b) **output/candidate_pairs.tsv** — not scored, but audited. It must be the **exact** candidate list that our matching model scores (the last blocking/filter stage), with header `source1_entity_id <TAB> candidate_entity_ids`. Every matched ID must also appear here. The organisers use it to measure our blocking recall and reduction ratio.

(c) **The final zip** — `<team_name>_submission.zip`:

```
<team_name>_submission.zip
├── output/
│   ├── matching_results.tsv
│   └── candidate_pairs.tsv
├── code/business_entity_resolution/
│   ├── src/           all source code
│   ├── README.md      exact steps: data → blocking → matching → output
│   └── requirements.txt pinned versions
└── Documentation_template.md  filled-in methodology (.md or .pdf)
```

The methodology document must cover the method, the blocking strategy, the model and features, and results with error analysis. There is no page limit, and depth is valued.

5. How we are scored

$F0.5 = 1.25 \cdot P \cdot R / (0.25 \cdot P + R)$, computed **for each Source 1 entity** and then **averaged over all of them** (macro average). Precision counts twice as much as recall. Truth empty and prediction empty = **1.0**. Truth empty and anything predicted = **0.0**. Truth non-empty and prediction empty = **0.0**.

Situation (true matches = k)	Prediction	P	R	F0.5
Problem-statement example, k = 2	2 right + 1 wrong	0.67	1.00	0.714
k = 3	2 right, 0 wrong	1.00	0.67	0.909
k = 3	3 right + 1 wrong	0.75	1.00	0.789
k = 5	only the 1 surest match	1.00	0.20	0.556
Singleton, k = 0	empty	—	—	1.000
Singleton, k = 0	any 1 match	0	—	0.000

Takeaway: a missing link costs less than a wrong link, but an empty answer for a real entity scores zero. The best strategy is to add matches in order of confidence and stop as soon as the next one is more likely to hurt than help. The public leaderboard uses only a subset of the test set. **Final ranking uses the private remainder**, so thresholds must not be tuned to the public score.

6. Constraints and rules

Area	Rule	Consequence if broken
Format	Tab-separated, exact header names, UTF-8; one row per test S1; only existing S2-/S3- IDs; no duplicate IDs in a list; no duplicate rows	Submission rejected (not scored)
Completeness	Every test S1 must appear, including France and singletons (empty field)	Rejected
Model	Final model must be MIT or Apache-2.0 licensed and ≤ 8 billion parameters	Fails license review
Fair play	No external data: no entity-resolution APIs, business registries, geocoding APIs , or internet augmentation	Immediate disqualification
Reproducibility	Zip must regenerate both outputs from the raw data with only the shipped code, README and requirements	Top teams audited; may be disqualified
Country	Country is an open set: no hard-coding, filtering or one-hot to {US, India}	France rows lost or mis-scored

License quick-guide (check each model card before use). OK: LightGBM (MIT), XGBoost and CatBoost (Apache-2.0), multilingual MiniLM and LaBSE (Apache-2.0), multilingual-e5 and bge-m3 (MIT), Qwen2.5-7B and Mistral-7B (Apache-2.0). **Not OK:** Llama, Gemma, any model over 8B, non-commercial licenses. Keep libraries permissive too (e.g. avoid GPL Unidecode).

Our hardware: i5-13500H (16 threads), 15.6 GB RAM (often under 1 GB free), no CUDA GPU, about 12 GB free disk. The current Python environment is missing rapidfuzz, lightgbm and polars, and torch is broken against numpy 2, so we need a clean virtual environment. Work one country at a time and in chunks, and never build the full cross product (India alone is about 0.8M × 4.7M pairs).

7. Do and don't

DO	DON'T
✓ Read with <code>sep='\t', dtype=str, quoting=QUOTE_NONE, keep_default_na=False</code> and assert row counts	✗ Call any external API or geocoder, or download business or address data
✓ Write with <code>sep='\t', index=False, lineterminator='\n'</code> (Windows defaults to CRLF)	✗ Hard-code, one-hot or filter countries to {US, India}
✓ Group by whatever country strings exist; never enumerate them	✗ Use Llama/Gemma or any model over 8B in the pipeline
✓ Put every test S1 in both files, including France and singletons	✗ Force a "best available" match: singletons need an empty answer
✓ Enforce one owner per S2/S3 record	✗ Chase the public leaderboard (it is only a subset)
✓ Tune thresholds for macro per-entity F0.5 on out-of-fold data	✗ Rely on ZIP or PIN codes (they are essentially absent)
✓ Keep <code>candidate_pairs.tsv</code> equal to the model's real inference set	✗ Trust train-validation scores for France without a margin
✓ Run the validator before every upload (sometimes with <code>--check-ids</code>)	✗ Write CSV, quoted fields, CRLF, or spaces after commas
✓ Pin versions, fix seeds, use relative paths	✗ Use entity ID numbers as a signal
✓ Disclose all rule tables, libraries and model licenses in the documentation	✗ Let pandas turn 'NA' names into NaN or merge quoted rows

8. How we will solve it

```
raw TSVs
→ [0] robust load + integrity asserts
→ [1] normalise: text clean, leetspeak fix, alias split (dba/fka/aka), legal-form extraction,
    domain → name, Indic → Latin transliteration, address parse (house no., street, city, state)
→ [2] blocking per country label: each S2/S3 record retrieves its top-K S1 candidates via
    rare-token keys + char-3gram TF-IDF kNN + address keys + concat-name key + embedding kNN
    ==> candidate_pairs.tsv
→ [3] pairwise features (name, address, rarity, rank/margin; country NOT used as a feature)
→ [4] LightGBM matcher (+ optional ≤8B MIT/Apache cross-encoder on the uncertain band)
→ [5] decision: one owner per S2/S3, runner-up margin, keep the prefix that maximises expected F0.5
    ==> matching_results.tsv → validator → upload
```

Validation: 5-fold GroupKFold by S1 over training, with blocking against the full S2/S3 pool so distractors are realistic. Metric re-implemented exactly (unit-tested on the 0.714 example). **Leave-one-country-out** (train US → test India, and the reverse) serves as a stand-in for France, since no French labels exist. Track the blocking recall ceiling (target ≥ 98% of true links).

Milestones

#	Milestone	Done when
M0	Environment, loader, metric, writer	Clean venv; row-count asserts; metric tests; an all-empty file passes the validator
M1	Rule baseline on the leaderboard	First SCORED submission; baseline logged
M2	Multi-pass blocking	$\geq 98\%$ pair recall on validation within RAM; candidate_pairs.tsv produced
M3	ML matcher (LightGBM)	Out-of-fold macro F0.5 clearly above M1
M4	Multilingual and France robustness	Learned transliteration; French rules; leave-one-country-out gap reduced
M5	Decision layer	Assignment + expected-F0.5 selection tuned out-of-fold
M6	Package	Zip reproduces end-to-end from a clean copy; docs filled; validator PASS

9. Pre-submission checklist

- ☐ 1,732,544 data rows plus header in both output files; header names exact
- ☐ Tabs only, UTF-8, LF line endings, no quotes, no spaces in ID lists
- ☐ Only S2-/S3- IDs that exist in the test files; no duplicates; every match is also a candidate
- ☐ `python utils/validate_submission.py --matching output/matching_results.tsv --candidate output/candidate_pairs.tsv --test-dir dataset/test` prints PASS
- ☐ Zip layout exactly as specified; README reproduces from scratch; requirements pinned
- ☐ Documentation_template.md filled in; licenses and rule tables disclosed; no external data used

10. Open questions

- Deadline and daily leaderboard submission limit (not stated in the materials).
- Team name and member list (needed for the zip and the documentation).
- Confirm hand-written normalisation tables and offline transliteration are acceptable (we will disclose them either way).
- Confirm unsupervised use of test text (e.g. IDF fitting) is acceptable.
- Does the MIT/Apache rule apply to every dependency? (We assume yes, to be safe.)
- Watch the linked explainer video; it is not included in the files and may add details.

Full detail: docs/PRD.md. Raw EDA numbers: docs/eda_stats.json.